Aliah University

End-Semester Examination - 2024

(For BCA. 2nd Semester)

Paper Name: Computer Organization **Full Marks: 80**
Paper Code: BCAUGMCC1202 **Time: 3 Hrs**

Group – A **(10X1=10)**

1. **Fill in the blanks–**
   a) CPU execution time = Instruction Count * ________ * Clock rate.
   b) Size of 1 cell in Byte addressable memories = ___________.
   c) Computer data buses are _____directional.
   d) Each division in cache memory (of fixed block size) is called ____________.
   e) For a One-address instruction, any ALU operation is performed in ____________.
   f) The following instruction - MOV 5(R1) is an example of______ addressing mode.
   g) $(4AA1C)_{16}$ is equivalent to ____________________ in Octal.
   h) Cache Memory works on Principle of _______________________.
   i) Program Counter holds __________ of next instruction to be executed.
   j) 1-way set Associative mapping is exactly same as __________ mapping.

Group – B
Answer any six of the following questions - **(6X5=30)**

2. Explain the Little-endian and Big-endian concepts for Byte ordering with the help of an example. [5]
3. Mention the names and the specific role of all special-purpose registers. [5]
4. What is BUS? Explain various types of BUS in Computer Architecture. [1+4]
5. What is "Hit ratio"? Draw a comparative discussion of various memory levels in computer memory hierarchy concept. (in short) [1+4]
6. Define Addressing modes. Explain any 4-memory based addressing modes with examples.
7. A Main memory of size 16 GB and a Cache Memory of size 32KB are divided into equal sized block and the Block size is 1KB. Mention number of bits require for performing – Direct mapping and 4-way set associative mapping. [5]
8. What is "overflow" in addition of 2's complemented numbers? Mention the conditions to check whether an overflow has occurred. Give examples to explain. [1+4]

16 GB = 16 × 2^{10} × 2^{10} Kb

Group – C
Answer any four of the following questions - **(10X4=40)**

9. a) What is Instruction cycle? Explain it with the help of a flowchart.
   b) Consider a hypothetical system, which has 8-bit instruction and 2-bit addresses. How many 2-addresesss instructions are possible? If there already exist 3 (Three) 2-address instruction and 5 (Five) 1-address instructions, then how many 0-address instructions are possible? [4+6]

3 2-ad
5 1-ad
0-ad

$\boxed{4\ 8 + 2 + 2}$

10. a) Multiply (-11) and (-5) by using Booth's Algorithm.

b) Represent a decimal number $(-19)_{10}$ in 8-bit Sign-magnitude, 1's complement and 2's complement form. [7+3]

11. a) Suppose number is using 16-bit format: the 1-bit sign bit, 8-bits for signed exponent, and 7-bits for the fractional part. The leading bit 1 is not stored (as it is always 1 for a normalized number i.e. implicit normalization. Then represent $(-53.5)_{10}$.

b) What is the average memory access time for a machine with a cache hit rate of 80% and cache access time of 5 ns and main memory access time of 100 ns when Hierarchical access memory organization is used? [6+4]

$(T_1+T_2)(H_2)(1-H_1)$

12. a) What are the various Input/ Output transfer modes available? Write a brief note on DMA. Prog IO, Interrupt

b) Draw a comparative discussion between Hardwired, Vertical and Horizontal Designing concepts of Control Unit. [5+5]

13. Consider a system with 4 cache lines, with following main memory block references –

9,7,53,32,9,43,18,53,9,56,34,1,32,9

Assume initially cache is empty and then identify hit ratios by using

(a) Direct Mapping Technique, D

(b) 2-way set Associative with FIFO Approach. $\frac{1}{13}$ [4+6]

16	8	4	2	1
	1	0	1	1
		1	0	1